

PRAJWAL BENEDICT NORMAN

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EDUCATION

B.Tech in Computer Science : Dehradun Institute of Technology, Dehradun
2028

Aug 2024 – May

GPA: 8.51 | Relevant coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Database Systems

TECHNICAL SKILLS

Languages: Go, TypeScript, Python, JavaScript, Java

Frontend: React.js, WebSockets, Recharts, Canvas API

Backend & APIs: Node.js, Express.js, REST APIs, gRPC, JWT / OAuth flows

Databases: PostgreSQL, MongoDB, MySQL

AI / ML: LlamaIndex, Pinecone, Gemini API, RAGAS

Infrastructure & Observability: Prometheus, Grafana, Docker, Git

Blockchain / Web3: Ethereum JSON-RPC, Web Crypto API

PROJECTS

QueryGuard : LLM Query Drift Detection Sidecar | [GitHub](#) | Go, Prometheus, Grafana, Docker

- Architected QueryGuard, a Go sidecar for LLM query drift detection using concurrent CUSUM and EWMA algorithms with fan-out/fan-in goroutine pipelines.
- Achieved 18% cosine similarity drop detection ($0.82 \rightarrow 0.64$) on off-topic queries with 0% false positives on stable baselines; CUSUM threshold tuned to balance sensitivity against alert fatigue.
- Exposed 10 real-time Prometheus metrics with Grafana dashboards; stateless sidecar design means each instance reports independently with no shared state or coordination overhead.
- Designed a generic HTTP adapter pattern decoupling the sidecar from any specific LLM backend: supports any REST API without configuration changes.

EthLink : Self-Custodial Ethereum Wallet | [GitHub](#) | [Live](#) | React, TypeScript, Node.js, PostgreSQL

- Designed and deployed a production cryptographic system across 14 REST API endpoints: AES-256 encryption on private keys, bcrypt on passwords and refresh tokens, and rate limiting (5–10 req/15 min) on sensitive routes.
- Implemented the complete wallet lifecycle: key pair generation, encrypted seed backup, transaction signing, and concurrent incoming/outgoing history aggregation across Mainnet, Sepolia, and Holesky; parallel fetch pattern halves history load time vs sequential calls.
- Architected a companion SDK with an OAuth-style redirect flow enabling dApps to trigger transaction signing without ever accessing user private keys: full custody boundary enforced at the API layer.

Self-Improving RAG Pipeline : AI-Integrated System | [GitHub](#) | Python, LlamaIndex, Pinecone, Gemini API, RAGAS

- Built a self-improving RAG pipeline (LlamaIndex + Pinecone + Gemini API) running an 18-config grid search across 3 chunking strategies, 2 embedding dimensions (768-d and 256-d), and 3 query rewrite variants; evaluated via weighted RAGAS composite scoring.
- Best config (semantic chunking + 768-d embeddings + single-step query expansion) achieved composite RAGAS score of 0.776; system persists optimal config and auto-reoptimizes on new ingestion or 7-day staleness trigger.

MCP Testing Platform : Developer Infrastructure | [GitHub](#) | TypeScript, Node.js, MCP SDK

- Designed and shipped a testing framework for Model Context Protocol servers supporting 2 transport protocols (stdio and HTTP Streamable): automated tool validation, protocol conformance checking, and a React-based test dashboard.
- Deployed a remote MCP test server on Render with end-to-end test coverage across stdio and HTTP Streamable transport layers.

OPEN SOURCE EXPERIENCE

GirlScript Summer of Code (GSSoC) : Open Source Contributor

2025

- Added Google Sign-In via Firebase to Vehigo's signup flow: integrated signInWithPopup, auto-fill for 4 input fields, conditional password field handling, and dual auth_provider routing to backend. (PR #158, fixes #127)
- Built profile photo upload/update for SentiLog-AI using Multer + Cloudinary: multipart/form-data handling, secure_url persistence to DB, 4 test cases covering success/failure/auth paths, and fixed production deployment errors. (PR #117)
- Integrated a rich text editor (react-quill-new) into SentiLog-AI's Journal page with full light/dark mode support: dynamic skin switching, custom content theming, and glassmorphism UI enhancements. (PR #35)